```
#Web Scraping Script- A reproducible Python script or notebook that extracts data from a chosen website or API.
```

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import time
```

```
url ="https://books.toscrape.com/catalogue/page-1.html"
```

```
def scrape_books(page_url):
    headers = {"User-Agent": "Mozilla/5.0"}

    response = requests.get(page_url, headers=headers)
    soup = BeautifulSoup(response.text, "html.parser")

    books = soup.find_all("article", class_="product_pod")

    book_data = []

    for book in books:
        title = book.h3.a["title"]

        price = book.find("p", class_="price_color").text
        price = price.replace("£", "")

        rating = book.find("p", class_="star-rating")["class"][1]

        link = book.h3.a["href"]
        full_link = "https://books.toscrape.com/catalogue/" + link

        book_data.append({"title": title,"price": price,"rating": rating,"link": full_link})

    return book_data
```

```
all_books = []

print("Starting Books Scraping")

for page in range(1, 51):
    url = f"https://books.toscrape.com/catalogue/page-{page}.html"
    data = scrape_books(url)
    all_books.extend(data)
    time.sleep(1)

print("Scraping completed!")
```

```
Starting Books Scraping
Scraping completed!
```

```
df_books = pd.DataFrame(all_books)
```

```
print("Total records scraped:", len(df_books))
df_books.head()
```

```
Total records scraped: 1000
```

	title	price	rating	link
0	A Light in the Attic	Â51.77	Three	https://books.toscrape.com/catalogue/a-light-i...
1	Tipping the Velvet	Â53.74	One	https://books.toscrape.com/catalogue/tipping-t...
2	Soumission	Â50.10	One	https://books.toscrape.com/catalogue/soumissio...
3	Sharp Objects	Â47.82	Four	https://books.toscrape.com/catalogue/sharp-obj...
4	Sapiens: A Brief History of Humankind	Â54.23	Five	https://books.toscrape.com/catalogue/sapiens-a...

Next steps: Generate code with df_books New interactive sheet

```
df_books.drop_duplicates(inplace=True)
df_books["price"] = df_books["price"].str.replace("Â", "", regex=False).astype(float)
```

```
if not df_books.empty:
    df_books.to_csv("books_data.csv", index=False, encoding="utf-8")
    print("Books CSV saved successfully!")
else:
    print("No data scraped")
```

```
Books CSV saved successfully!
```